

Yang Zheng

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Education

New York University, Tandon School of Engineering

Expected May 2027

M.S. in Computer Science

Relevant Coursework: Cloud Computing, Big Data, Software Engineering, Computer Networking, Applied Cryptography, Blockchain & DLT

New York University, College of Arts & Science

2021 - 2025

B.A. in Computer Science & Mathematics

Experience

Teaching Assistant — Algorithms | NYU Courant Institute of Mathematical Sciences

May – Aug 2025

- Held weekly office hours and review sessions for CSCI-UA 310 undergraduates, fielding questions on mathematical proofs of algorithm correctness, LeetCode-style problem solving, and final exam preparation.
- Guided students through divide-and-conquer, greedy algorithms, dynamic programming, graph algorithms, and NP-completeness; reinforced rigorous proof techniques (induction, exchange arguments, reductions).

Research Assistant | Robotics and Automation Laboratory, Tsinghua University

June – Aug 2023

- Developed a data-driven method to estimate 3-D states of deformable linear objects from single-frame occluded point clouds, addressing real-world occlusion challenges in robotic perception.
- Trained a two-branch neural network exploiting global and local point cloud information; designed a fusion module combining both approaches for improved 3-D state estimation accuracy.

Projects

MiniVault — Crypto-Collateralized Stablecoin | Solidity, TypeScript, React, Hardhat

Spring 2026

Ethereum (Sepolia), Chainlink, OpenZeppelin ERC-20, ethers.js v6, TypeChain, Tailwind, Vite

- Designed and deployed three Etherscan-verified contracts (MiniUSD, OracleAdapter, Vault) on Ethereum Sepolia supporting deposit, mint, repay, withdraw, and liquidation against a live Chainlink ETH/USD feed at 150% collateralization.
- Engineered a same-block oracle circuit breaker halting minting on >5% intra-block price divergence (cf. Mango Markets, 2022); validated with a 33-case Hardhat test suite and a React + ethers.js frontend.

Research Paper Summarizer | TypeScript, AWS CDK, Next.js, Amazon Bedrock

Spring 2026

AWS Lambda, API Gateway, Step Functions, SQS, DynamoDB, S3, CloudFront, Cognito, CloudWatch, X-Ray

- Owned the Search Lambda on a 4-person team — an arXiv API client with parallel querying, paging, retry, and Cognito-authorized API Gateway routing, deduplicated with Semantic Scholar and tested in vitest.
- Built the ops layer (CloudWatch dashboard, SNS alarms, X-Ray tracing across the Step Functions pipeline); helped land a v1.0 release at ~30s per summary, ~\$0.20 per paper, \$0 idle.

Dining Concierge Chatbot | Python, JavaScript

Spring 2026

AWS S3, API Gateway, Lambda, Lex, SQS, DynamoDB, OpenSearch, SES, Yelp Fusion API

- Architected a serverless pipeline across 9 AWS services delivering personalized restaurant recommendations from 1,300+ Manhattan restaurants, with DynamoDB-backed state persistence across 5 conversational intents so returning users could retrieve previous dining preferences.

Programming Language Interpreter | Scala

Spring 2025

- Implemented a full interpreter for a subset of JavaScript including lexing, parsing, and evaluation, supporting closures, higher-order functions, and recursive data structures.

Graphics Animations and Programs | JavaScript, WebGL, OpenGL

Fall 2024

- Built ray tracing and rasterization renderers, real-time human-motion animations, and a textured solar system simulation.

Technical Skills

Languages: C++, Python, Java, JavaScript, TypeScript, C, SQL, R, Scala, Solidity, HTML/CSS

Frameworks & Tools: React, Next.js, Node.js, MongoDB, PyTorch, WebGL, OpenGL, Tailwind CSS, Bootstrap, Hardhat, ethers.js, Git

Cloud & Infrastructure: AWS (Lambda, API Gateway, Step Functions, SQS, S3, DynamoDB, OpenSearch, CloudFront, Cognito, Bedrock, CloudWatch, X-Ray, Lex, SES), AWS CDK, Serverless Architecture